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- **Term:** 3 AY25-26

Required Python Packages

Run the following applications once if they are not yet installed in your Python environment:

```
In [1]: !pip install numpy
```

```
'pip' is not recognized as an internal or external command,  
operable program or batch file.
```

```
In [2]: !pip install scipy
```

```
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operable program or batch file.
```

```
In [3]: !pip install matplotlib
```

```
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operable program or batch file.
```

```
In [4]: !pip install panda
```

```
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operable program or batch file.
```

```
In [5]: !pip install gekko
```

```
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operable program or batch file.
```

```
In [6]: !pip install scikit-learn
```

```
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operable program or batch file.
```

```
In [7]: !pip install sympy
```

```
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operable program or batch file.
```

Machine Problem 002: Inverse Laplace Transforms

Finding a function $f(t)$, given the Laplace transform $F(s)$. If $\mathcal{L}f(t) = F(s)$ we write:

$$f(t) = \mathcal{L}^{-1}(F(s))$$

The SymPy library is a symbolic mathematics library written for Python that can be used to perform analytical calculations. This library can be very useful for control system design and analysis.

Table of Common Laplace Transforms

Mathematicians have developed tables of commonly used Laplace transforms. Here is summary table with a few of the entries that will be most common for analysis of linear differential equations.

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin(\omega t)u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos(\omega t)u(t)$	$\frac{s}{s^2 + \omega^2}$

```
In [8]: import sympy as sym
from sympy.abc import s,t,u,x,y,z,a,b,c,k
from sympy.integrals import inverse_laplace_transform
```

```
In [9]: # For pretty printing
sym.init_printing()
```

```
In [10]: # Define variables for time 't' and complex variable 's'
# t = sym.symbols('t', real=True)
# s = sym.symbols('s')

# Define a constant:
# a = sym.symbols('a', real=True, positive=True)
```

1. Evaluate:

$$\mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} = t^2$$

```
In [11]: # Define the frequency domain function:
f1s = 2/s**3
f1s
```

Out[11]: $\frac{2}{s^3}$

```
In [12]: # Compute the Inverse Laplace Transform:
f1 = inverse_laplace_transform(f1s, s, t)
f1
```

Out[12]: $t^2\theta(t)$

2. Evaluate:

$$\mathcal{L}^{-1}\frac{s}{s^2+1} = \cos(t)$$

```
In [13]: # Define the frequency domain function:
f2s = s/(s**2+1)
f2s
```

Out[13]: $\frac{s}{s^2+1}$

```
In [14]: # Compute the Inverse Laplace Transform:
f2 = inverse_laplace_transform(f2s, s, t)
f2
```

Out[14]: $\cos(t)\theta(t)$

3. Evaluate:

$$\mathcal{L}^{-1}\frac{1}{s^2+1} = \sin(t)$$

```
In [15]: # Define the frequency domain function:
f3s = 1/(s**2+1)
f3s
```

Out[15]: $\frac{1}{s^2+1}$

```
In [16]: # Laplace transform (t->s)
f3 = inverse_laplace_transform(f3s, s, t)
f3
```

Out[16]: $\sin(t)\theta(t)$

4. Evaluate:

$$\mathcal{L}^{-1} \frac{10}{(s+2)^4} = \frac{5t^3 e^{-2t}}{3}$$

In [17]: *# Define the frequency domain function:*

In [18]: *# Compute the Inverse Laplace Transform:*

5. Evaluate:

$$\mathcal{L}^{-1} \frac{s+1}{(s+1)^2+4} = e^{-t} \cos 2t$$

In [19]: *# Define the frequency domain function:*

In [20]: *# Compute the Inverse Laplace Transform:*

6. Evaluate:

$$\mathcal{L}^{-1} \frac{15}{(s-1)^2-9} = 5e^t \sinh 3t$$

In [21]: *# Define the frequency domain function:*

In [22]: *# Compute the Inverse Laplace Transform:*

7. Evaluate:

$$\mathcal{L}^{-1} \frac{4s-1}{s^2-s} = 1 + 3e^t$$

In [23]: *# Define the frequency domain function:*

In [24]: *# Compute the Inverse Laplace Transform:*